

SCHEDULE

Thursday, April 27, 1995 (LSC 4258/63)

9:00 - 9:15 Coffee

9:15 - 9:30 Opening Remarks—Richard Brown

9:30 - 11:00 Session 1. Chair: Tara Fidler

- T1) H. Piggins & B. Rusak
- T2) D. Phillips, M. Semple, L.M. Kitze, & S.E. Hall
- T3) G. Scott & B. Rusak
- T4) S. McColl
- T5) K. Forbes & R. Klein
- T6) L. Lambrecht

11:00 - 11:15 Coffee Break

11:15 - 12:30 Session 2. Chair: Heather Schellinck

- T7) S. Gadbois, W. Moger, P.J. McLeod, C.J. Ryon, & J. Fentress
- T8) J. Fentress & V. Bolivar
- T9) K. Semba, L. Detari & D. Rasmusson
- T10) R. Brown
- T11) S. Stewart & S. Samoluk

12:30 - 2:00 Lunch Break and Posters (Lounge):

- P1) L. Lambrecht
- P2) M. Guido, B. Rusak & H. Robertson
- P3) S. McColl & W. Schmidt
- P4) F. Dastur, R. Brown, P. McGrath, D. Brown, K. Harrigan & H. Wiles
- P5) C. MacKinnon & W.G. Tatton

2:00 - 3:30 Session 3. Chair: Brad Frankland

- T12) W. K. Honig & B. Matheson
- T13) P. McMullen
- T14) W. Schmidt
- T15) J. Christie
- T16) T. Fidler & V. LoLordo
- T17) S. Shaw & J. A. Horne

3:30 - 3:45 Coffee Break

3:45 - 5:00 Session 4. Chair: Kelly Forbes

- T18) C. Moore
- T19) J. Barresi, C. Selig & C. Moore
- T20) M. MacGregor & K. Davidson
- T21) A. MacDonald, S. Stewart, S. Samoluk & S. Sweet
- T22) B. Frankland & A. Cohen

There will be wine and cheese served in the Lounge after the last session of the day.

Friday, April 28, 1995 (LSC 4258/63)

9:00 - 9:15 Coffee

9:15 - 10:45 Session 5. Chair: Tim Juckes

T23) K. Davidson, Y. Gidron (Speaker) & B. Chaplin

T24) R. Hoffman

T25) B. Losier, C. Bilsbury & A. Richman

T26) B. Losier

T27) M. MacGregor & K. Davidson

T28) T. Taylor, D. Phillips & S.E. Hall

10:45 - 11:00 Coffee Break

11:00 - 12:30 Special Speaker: Dr. Ellen Bialystok, Department of Psychology, York University
- introduced by ?
"What the Squiggles Mean: Children's Understanding of Symbolic Notation"

12:30 - 2:00 Lunch Break and Posters (Lounge):

P6) J. Hamm & P. McMullen

P7) F. Dastur, R. Brown, D. Wiles & I. McGregor

P8) B. Chaplin

P9) B. Matheson, S. Gadbois & W.K. Honig

2:00 - 2:45 Session 6. Chair: Michael MacGregor

T29) M. Watt & J. McGlone

T30) H. Schellinck & R. Brown

T31) A. Israeli & G. Eskes

2:45 - 3:00 Coffee Break

3:00 - 4:45 Symposium "Where Are We Going? Psychology and Neuroscience in the 21st Century"

Chair: Richard Brown ~~300~~ 300

• Jim Clark ~~305~~ 315

• Bill Tatton ~~305~~ 330

• John Fentress ~~340~~ 345

• Ray Klein ~~35~~ 400

• John Barresi 415

• Pat McGrath 430

4:45 - 5:00 Closing Remarks and Awards - title / data point

There will be pizza and beer served in the Lounge after the last session of the day.

All talk sessions will be held in the LSC Rooms 4258/63.

T1. Getting a GRiP on your Very Important Parts

Hugh Piggins and Ben Rusak

The mammalian circadian pacemaker is found in the suprachiasmatic nucleus (SCN). The SCN contain a number of neuropeptides including gastrin-releasing peptide (GRP), peptide histidine isoleucine (PHI), and vasoactive intestinal polypeptide (VIP). We tested a model which predicts that these neuropeptides interact synergistically to exert a maximal effect on the circadian organization of behaviour. GRP, PHI, and VIP, alone or in a neuropeptide cocktail (GRP+PHI+VIP; shaken, not stirred), were injected into the SCN of Syrian hamsters. We found that each neuropeptide could alter the circadian organization of behaviour. The cocktail also had large effects, but these were less than the additive effects of each neuropeptide alone. These data do not support the hypothesis that these neuropeptides interact synergistically to exert maximal effects on the the circadian organization of behaviour.

T2. Once Is Not Enough: Turning Your Partner On With Multiple Spikes

D.P. Phillips, M.N. Semple, L.M. Kitzes and S.E. Hall

In independent studies, Phillips (Dalhousie) and Eggermont (Calgary) are compiling circumstantial evidence on local circuits in the cerebral cortex. Phillips & Sark (1991) reported that many cells in the cat's primary auditory cortex (AI) had responses consisting of a burst of spikes; the inter-spike intervals in these responses were very short (≈ 2 ms) and were typically stimulus-invariant. Bowman, Eggermont & Smith (1995, submitted) confirmed and extended these findings. They suggested that most extracellular recordings in AI come from larger neurons (likely pyramidal and basket cells), while the smaller (nonpyramidal) cells are relatively inaccessible to metal electrodes. There is reason to believe that the small cells are probably inhibitory interneurons that receive input from the pyramidal cells, and which, either through post-synaptic membrane specializations or local GABA-mediated inhibition (Winer, personal communication), are best excited by a train of closely-spaced pyramidal cell spikes. If these pyramidal cell/nonpyramidal cell circuits are a property of the cerebral cortex generally, rather than a property specifically of AI, then the same phenomenology should be seen in other cortical areas. We have obtained evidence on the responses of neurons in the posterior auditory field, studied under the same conditions as our earlier studies of AI. Can you guess what we've found?

T3. Histamine Activates / Inhibits / Modulates / Has No Effect / Does Whatever I Want It To* In the SCN (*Delete As Appropriate)

George Scott & Benjamin Rusak

us found?
ind
messages?
These studies investigated the effects of histamine on spontaneous and NMDA-evoked activity in the hamster SCN in vitro. Application of histamine either activated (n=13/27), attenuated (n=6/27) or had no effect (n=8/27) on spontaneous firing in SCN cells. Attenuation of spontaneous firing rate by histamine was evident at higher ejection currents than those which elicited activations. Application of NMDA increased the firing rates of single SCN cells in a current-dependent manner. Co-application of histamine, at ejection currents which elicited no direct response to histamine itself, potentiated NMDA-evoked activity by up to 40% (61% of cells tested). These results suggest that, in addition to having direct effects on spontaneous neuronal activity in the SCN, histamine may modulate the activity of NMDA receptors at glutamatergic synapses in the hamster SCN. And a bunch of other stuff. *A molecular switch?*

T4. Interocular Transfer Of Motion Adaptation Effects: Higher-Order Processing of Motion Stimuli In Normal and Stereoanomalous Observers

Shelley McColl

Interocular transfer (IOT) of conventional motion aftereffects (MAE's) has been employed as a psychophysical technique to assess the binocularity of the human visual cortex. Like IOT of other aftereffects, conventional MAE's show partial IOT (approx 60%). Recently, Raymond (1993) observed that with a higher-level motion coherence after-effect, humans with normal stereopsis show complete IOT. Her argument is that conventional MAE's probe lower-level visual areas, where both monocular and binocular cells exist, whereas higher-order MAE's tap into higher-order visual areas, where cells are almost exclusively binocular. In view of the fact that the cortical area MT (in monkeys), is thought to mediate coherent motion, and is more binocular than area 17, the present experiment will revisit the issue of how stereoabilities are related to IOT using Raymond's (1993) higher-order MAE task.

T5. The Magnitude of the Fixation Offset Effect with Endogenously and Exogenously Controlled Saccades

Kelly Forbes and Raymond Klein

Two experiments examined saccadic reaction time (RT) as a function of fixation offset condition (no offset; simultaneous and 200 ms offset-target SOA) in prosaccade and antisaccade tasks. So that observers would be equally prepared in each condition auditory warning tones preceded target onset on every trial. The second experiment also included saccades in response to verbal commands. The fixation offset (or gap) effect (FOE) was identical in the prosaccade and verbal tasks. The FOE in the antisaccade task was significant but it was significantly smaller than in the other tasks. We speculate that the reduced FOE in the antisaccade task may be due to the requirement to inhibit the superior colliculus when the target directed saccadic programs are, by instruction, erroneous.

T6. Two Year Olds' Conversation With a Robot

Linda Lambrecht

end session!

T7. Social Stress In Captive Timber Wolves (*Canis lupus*): Individual Differences In Cortisol Levels

S. Gadbois, W. Moger, P.J. McLeod, C.J. Ryon and J.C. Fentress

Urine samples from fourteen known individuals within a captive timber wolf pack were collected during the breeding season from snow and analyzed for levels of stress hormone. Social behavior during this period was also recorded. Individual differences were found in urinary cortisol concentrations (expressed as cortisol:creatinine ratio, correcting for differences in urine concentration and dilution in snow). The lowest ranking female had a significantly elevated cortisol:creatinine (c:c) ratio than other females and the second (beta) ranking male, whose social position was less stable, had significantly higher c:c ratios than most other males. The data presented also suggest c:c levels increase during the course of the breeding season and may vary in response to even short periods of high social stress. Similar ratios have been reported from domestic dogs and wild wolves, suggesting that data from the present sample of captive wolves are not abnormal.

T8. Do Mutant Mice Make Sense?

John Fentress and Valerie Bolivar

We think so.
But only if evaluated in detail.
So, we designed some tests.
These tests involve careful developmental analyses of motor development.
They are really cool.
So, you should listen to us tell you about them.
For example: How do you sort out the compromising effects of different brain regions in development and motor performance? Or, how do you think about single gene effects on single brain regions or single aspects of motor performance? Or, how do you think about network properties in behaviour?
Maybe mutant mice can help us here.
But we have to think carefully.
So, why study grooming in weaver mice? Come by, and we will tell you!

T9. Synaptic Control of Wake- and Sleep-Active Basal Forebrain Neurons: An Update

Kazuo Semba, Laszlo Detari (Anatomy and Neurobiology) & Douglas Rasmusson (Physiology and Biophysics)

Last year we presented a series of working hypotheses to examine the role of basal forebrain neurons in cortical arousal and, in particular their synaptic inputs ascending from the brainstem. We (Laszlo) have since done experiments and some of the data with brainstem stimulation were rather unexpected, as will be discussed. These data nevertheless support the hypothesis that basal forebrain neurons receive strong activation from the brainstem and provide in turn a major excitatory drive to the cerebral cortex.

- T10. NMDA receptors, MK -801, C-fos, DNA, mRNA, Protein Synthesis, Northern Blots, Polymerase Chain Reactions, Antisense Oligonucleotides and Behaviour. Is this the future of Psychology?**

Richard Brown

A few years ago, I read a paper entitled "Basic concepts of molecular biology for the reproductive endocrinologist" (Plouffe, L. 1991. Seminars in Reprod. Endocrinol. 9, 1-13) and thought that a paper entitled "Basic concepts of molecular biology for the experimental and clinical psychologist" might be useful. Lately, we have been working on a grant proposal entitled "Molecular mechanisms underlying odour-induced puberty in mice" and I have read a paper entitled "Antisense oligonucleotides: A new tool in neuroscience" (Pilowsky, et al., 1994, Clinical and Exp. Pharmacol. Physiol., 21, 935-944). Should there be a similar paper entitled "Antisense oligonucleotides: A new tool in experimental and clinical psychology"? My talk will focus on my struggle to understand the new molecular biology approaches to the study of behavioural neuroscience, with reference to our proposed research on odour-induced puberty. It will, no doubt, bear on some of the topics to be discussed in the Symposium on Future Directions in Psychology.

- T11. "I Don't Drink Very Often, But When I Do, I Drink A LOT!": Self-Reported Alcohol Consumption Levels and Selective Processing of Alcohol Cues Among Dietary Restrictors**

Sherry Stewart and Sarah Barton Samoluk

Previous research suggests important relationships between restrained eating and alcohol use/abuse in non-clinical samples. However, previous studies have failed to separately assess quantity and frequency measures of typical alcohol consumption, so that it remains unclear whether the observed positive relationship between dietary restraint and typical alcohol consumption levels is due to an increased quantity and/or frequency of alcohol consumption among highly restrained eaters. Moreover, previous research has demonstrated that non-clinical samples of restrained eaters demonstrate a cognitive bias favouring the selective processing of food cues. Whether this bias extends to the selective processing of alcohol cues remains unexplored. A sample of university students (78% F) was divided into three dietary restraint groups based on scores on the Restraint Scale. Subjects completed a reliable and valid measure of alcohol consumption, which assesses typical frequency of drinking episodes per week, and typical number of alcoholic drink equivalents consumed per drinking occasion. Subjects then completed a modified version of the Stroop task in which they colour-named balanced sets of food, alcohol, and neutral control words. Consistent with previous results, high restraint subjects reported drinking significantly more alcoholic beverages on a weekly basis than low restraint subjects. However, although high restraint subjects reported drinking significantly more alcoholic beverages per drinking occasion than low restraint subjects, there were no significant group differences in reported drinking frequency. Thus, high restraint subjects appear to drink no more frequently than others, but when they do drink, they drink significantly greater quantities of alcohol. Also consistent with previous research high, but not low or moderate, restraint subjects required significantly longer to colour-name food vs. neutral control words. High, but not low or moderate, restraint subjects also required significantly longer to colour-name alcohol vs. neutral control words. Thus, high restraint subjects displayed a cognitive bias favouring the processing of alcohol cues as well as food cues. Implications of the results for risk for abusive drinking among high restraint university students will be discussed.

and session 2

T12. Evidence For Transfer Between Spatial Locations And Their Pictured Representations In The Pigeon

W. K. Honig (presenter) and W. R. Matheson

In a rectangular room, three groups of pigeons first discriminated the end in which a baited feeder was located. The birds then discriminated between pictures of the ends of the room. For group AAB, this picture discrimination was congruent with initial training; for Groups ABA and ABB, it was incongruent. In the third phase, Group ABA found food in the initially baited feeder, while Groups AAB and ABB found food in the initially unbaited feeder. The latter groups learned the picture discrimination in the second phase more slowly than AAB, indicating transfer effects from room training to picture discrimination, which had not previously been obtained by Cole and Honig (1994). These data suggest that experience with nonreinforcement during the first phase influenced the degree of positive or negative transfer between the spatial and picture discriminations. In the third phase, all groups acquired the second feeder discrimination with few errors.

T13. How Relevant are Orientation-Specific Templates to Identification of Real-World, Rotated Objects?

Patricia A. McMullen, Dept. of Psychology, Dalhousie University

The time to name a set of objects rotated in the picture-plane increases monotonically with object orientation. This effect supports normalization of object images to the upright. However, it diminishes as the objects are repeatedly named in different orientations. Using a set of visually-similar, nonsense objects, Tarr and Pinker (1989) demonstrated that this effect diminishes because orientation-specific templates are formed each time the objects are identified. Consequently, object images have decreasing angular distance to normalize to the nearest orientation-specific template over successive identifications. However, Murray, Jolicoeur, McMullen & Ingleton (1993) failed to support this view for real-object naming. Are their results discrepant with those of Tarr & Pinker because orientation-specific template formation relies on identification of nonsense objects, or visually-similar objects? In new experiments reported here, Tarr & Pinker's findings were replicated when their objects were identified. However, no support for the formation of orientation-specific templates was found in two other experiments, one in which a set of visually-dissimilar nonsense objects was identified, and another in which a set of visually-similar real objects were identified. These findings undermine the relevance of orientation-specific template formation to object identification. However, human faces are probably the most visually-similar stimuli we normally identify. In a final experiment designed to optimize the formation of orientation-specific templates using real-world stimuli, a set of human faces is being identified.

T14. Now You See It, Now You... See It Again? The Two Motion Percepts Illusion

William C. Schmidt

If attention is exogenously cued, with a small lead time, to the location corresponding to one end of a quickly drawn line, then illusory line motion results with the percept that the line was drawn away from the cued location (Hikosaka, Miyauchi, & Shimojo, 1991). The current paper reports a similar illusion which replaces the line rendering with a dot moving at high velocity toward (or away from) the cued location. This method yields two percepts of motion, hence it is called the "Two Motion Percepts (TMP) illusion". The dot is (potentially) incorrectly perceived as travelling away from the locus of attention and then to double back and travel toward the locus of attention. The illusion was investigated through three experiments. Experiment 1 reports on the perceptions of 35 undergraduates when presented with the TMP displays. Over ninety percent of individuals reported the illusion. An extended version of Hikosaka et al.'s gradient model was proposed to account for both illusions. An additional assumption was required that attention not only speeds the initial transmission of signals, but that it also temporally extends signal transmission. Five predictions of the model were tested with eight observers in Experiment 2. All five predictions of the model were verified, and the data suggested that the effects of attentional cuing covered a large spatial area. Experiment 3, a control experiment with three observers, verified that cuing attention to a spatial location can affect stimuli as far as 7° away. Some implications of the extended model are discussed.

T15. Space, The Final Frontier (Pavlovian). A Study Designed To Boldly Go Where No One Has Gone Before

John Christie

There has been no research that I am aware of that has tested whether spatial contiguity of CS and US--unconfounded by delay of rft--influences Pavlovian conditioning. A unique box was designed that allowed CS-US distance to vary without varying CR-US interval (delay of rft). Four ring necked doves were autoshaped in this apparatus. Spatial contiguity of CS and US had no detectable effect.



T16. Shock Induced Finickiness: A Preliminary Result

Tara L. Fidler and Vincent M. LoLordo

There are many reports in the literature of finickiness in rats which have been shocked but not in rats which have not been shocked. Finickiness is a suppression of consumption of for example a bitter tasting solution relative to baseline consumption levels. In studies reported previously (for example see Job and Barnes, 1995, *Behavioral Neuroscience*, 109, 106-116) the finickiness effect has been reported only following the sorts of shock parameters that result in learned helplessness, 100+ shocks of 2-60s in duration. Previous work done in our lab had incorporated a one hour shock session of 60 2-s shocks and had failed to show a relationship between shock and alcohol consumption. In order to address the possibility that this failure resulted from a failure to produce differences in level of stress or fear between our shock and no-shock rats we sought independent evidence that animals shocked in our experiments experienced more "stress" than animals which were not shocked. Using our shock parameters in a finickiness experiment we showed that shocked rats did suppress their consumption of quinine relative to baseline levels of water consumption while no-shock rats did not.

T17. Stop Waving The Frogs' Legs, Just Pass Me The Toast

Steve Shaw and Jane Anne Horne

Proximity to high voltage transmission lines and to some electrical equipment can expose the body to low-frequency electromagnetic fields (EMFs). While EMFs are reputedly able to interact with biological tissue, the effects are expected to range from mild to non-existent at the field strengths normally encountered by most of us, although there is some controversy about this. An exception to the supposed subtlety of action is a sledgehammer effect reported five years ago by two British anaesthesiologists using a well-known preparation, the frog's sciatic nerve. They claimed that a short exposure to EMFs generated by a small coil surrounding the nerve is sufficient to disable it for several minutes, after which transmission recovers fully. They attribute this result to direct action by the applied EMF on some component of the nerve, which leads to what they term "electromagnetic inhibition" (EMI). The mechanism of action has not been investigated, but might be due to field-induced changes in the distribution of ions in or around the axons, they say.

Sensing a sizeable rodent here while on the look-out for more modern material to incorporate into a teaching lab, we have attempted to retrace the footsteps of these pioneers. We are able to reproduce what looks like their basic EMI phenomenon, although it is not possible to duplicate the exact conditions they used originally because too little detail is given in the published methods. Our preliminary measurements have led us to reject their explanation of EMI and to propose a rival theory, unveiled in public for the first time here: the Electromagnetic Toaster (EMT) hypothesis of nerve inhibition.

end session 3

T18. Conflicting Desire And The Child's Theory Of Mind

Chris Moore

We contrast the standard representational theory of mind approach to the understanding of mental states with an alternative view which has it that theory of mind tasks require executive functioning or the inhibition of more 'cognitively salient' information. In the first experiment 3-5-year-olds were tested on their understanding of desire in others either when they themselves held a strong and conflicting desire or when they had no strong desire. Results showed that under the condition of having a strong and conflicting desire, only 5-year-olds were able to recognize that another person might desire something different. In contrast when the children themselves held no strong desire, even 3-year-olds were able to judge another's desire correctly. The second experiment compared 3-year-olds' performance on a standard false belief task with that on an equivalently structured desire task in which subjects had again to inhibit their own strong and conflicting desire. Results showed similar performance on the traditional false belief task and the new conflicting desire task.

T19. Beyond Immediate Self-Interest: Or, What Good Comes From Having A Theory of Mind?

John Barresi, Carol Selig and Chris Moore

The purpose of this research was to investigate the early development of prudence and altruism involving future situations. The study tested the hypothesis that these are interrelated phenomena and that both should develop at approximately four years of age. Using a modified delay of gratification paradigm, 3- to 5-year-olds were tested on their ability to forgo a current desire for some stickers in order to further their own future interests or the current or future interests of a research assistant. Results showed that in current situations, altruism was unrelated to age. However, prudence and altruism involving the future situations were correlated with one another and with age. Children under four years of age performed significantly less well than the older children on the prudence task ($F(1, 49) = 15.75; p < .001$) and on the altruism task involving future situations, ($F(1, 49) = 33.24; p < .001$). Between three and five years, prudence and altruism showed a similar developmental pattern with performance increasing at age four.

T20. In Defense of Defenses: A User's Guide to the Psyche

Michael Wm. MacGregor and Karina Davidson

Two studies were conducted to determine coder reliability and the reliability of 25 defense mechanisms assessed using the Defense-Q, an idiographic q-sort measure. In the first, study 30 subjects who had undergone an interpersonally stressful structured interview were rated by ten trained coders for the use of 25 defense mechanism. Study two was similar to study one except that a new set of thirty interviews were rated by ten newly trained coders. The average coder reliability was .64 in study one (range .53 to .67) and .75 in study two (range .63 to .76) across all 25 defense mechanisms. Reliability coefficients for each defense mechanism ranged from .15 (reaction formation and displacement) to .87 (humour and acting out) in study one, and from .28 (undoing) to .92 (humour) in study two. These results indicate that defense mechanisms can be reliably assessed using a standardized clinician-guided q-sort technique. The applications of these results to the assessment of normal and abnormal personality will be discussed.

How people cope with stress at work, in every day life

T27

regression
denial

fantasy

humor

identification

projection - ego

projection

regression

projection

T21. Gender Differences In Consistency of Alcohol Consumption Measures

Alan MacDonald, Sherry Stewart, Sarah Samoluk, Susan Sweet

Kidorf et al. (1990) found that the self-reports of males were highly predictive of their analogue alcohol consumption, whereas the self-reports of females bore no relation to their analogue consumption. The present study examined the effects of social context on the consistency between self-report and analogue alcohol consumption in males vs females.

Male and female university students, assigned to either a social or alone drinking context, provided self-reports of alcohol consumption, and then participated in an analogue drinking test disguised as a beverage taste-rating task. Alcohol consumption dependent measures were analyzed as a function of context and gender. Preliminary correlations show that self-reports were more consistent with analogue drinking in both males and females in the social vs the alone context. Consistency between analogue drinking and self report was approximately equal for males and females in both contexts. Possible reasons for differences between these results and those reported by Kidorf et al. (1990) are discussed.

T22. Boundary Formation and Melodic Memory

B. W. Frankland & A. J. Cohen

Most people can recognize a large repertoire of melodies. The present two-part study examines how their melodic information is stored. In particular, it focuses on the role of boundary formation. It first addresses the issue of determining where boundaries arise. Secondly, it examines how boundaries at different positions in a melodic excerpt affect memory. Following comparable work in psycholinguistics (Jarvella, 1970; Wingfield & Butterworth, 1984), it was hypothesized that excerpts not straddling a boundary would be more easily recognized than excerpts which straddle a boundary. In Part 1 of the Experiment, 20 subjects listened to a familiar melody and depressed a key each time they detected a boundary. Note-by-note boundary profiles generated for each subject were significantly intercorrelated (with the exception of one subject). Group profiles were highly consistent across three presentations of the song. In Part 2, subjects heard 16 four-note excerpts from the song (50% contained an error) and judged whether the excerpt belonged to the melody. Generally, excerpts which straddled the boundary led to poorer performance. Excerpts that did not contain boundaries and excerpts that ended with a boundary led to higher performance. Thus, the memory structure for melodies entails a systematic boundary formation process not unlike that found for language. While this proposal is not new for music, the current methodology permits its quantitative investigation by the use of boundary assessment and subsequent memory testing that is tailored to the individual subjects within the study.

end session 4

T23. The Statistical Versus Clinical Significance of Medication, Psychotherapeutic and Cardiac-Rehabilitation Trials with Post-MI Patients

Karina Davidson, Yori Gidron and Bill Chaplin

This study compared the methodology and efficacy of Medication, Psychological and Mixed (cardiac-rehabilitation) trials with post-myocardial-infarction (MI) patients, using meaningful effect sizes. Twenty-four randomly selected and representative trials (eight per each trial type) were compared regarding prevention of reinfarcts and mortality. Tests of statistical significance (X²) and clinical significance (number of prevented reinfarcts or deaths in treatment group per 100 people), and a comprehensive methodological assessment scale were employed. Medication trials obtained statistically significant effects at the highest level. However, Psychological trials prevented 10 times more reinfarcts (2.9) and three times more mortalities (1.7) per 100 people than did Medication trials (.3, .5, respectively). Medication trials obtained higher mean quality ratings (38) than Psychological (31) and Mixed trials (33). Thus, tests of statistical significance may be misleading when comparing clinical trials with different sample sizes. Psychological trials should increase their sample sizes and improve their methodology to reveal their stronger clinical significance with greater reliability and validity.

T24. Caught in the Web: Psychology on the Internet in 1995

Ron Hoffman

The past year has seen a tremendous amount of activity on the World Wide Web, currently one of the most popular segments of the Internet. Individuals, businesses, governments and institutions of all sorts are making information about themselves available electronically, taking advantage of the hypertext and graphic capabilities of web browser software. Today it is possible to tour the Louvre Museum, book a hotel room in San Francisco (and view the room, of course), or order fresh Nova Scotia lobsters delivered anywhere in North America, all from your computer workstation. Tomorrow even more will be possible. We should be taking advantage of this medium, and in this presentation, I will demonstrate a Web site (Home Page) for the Dalhousie Psychology Department.

WWW

Mosaic - Web Browser

Netscape -

Home page = entry page to a web site.

T25. The Justification For Depression Scales: A Bibliometric Analysis

Bruno J. Losier, C.D. Bilsbury & A. Richman

Depression is a topic of increasing importance in both research and therapy. Its measurement has for three decades depended largely on two rating scales, the Hamilton and the Beck. These warhorses vie for acceptance as the gold standard. Is there justification for their preeminence? This bibliometric analysis begins to answer this question. What are the time trends in citations? Are the Hamilton and Beck co-cited? Is the selection of one justified over the other? When both scales are used, are the results compared? We have used PSYCLIT and Science Citation Index to count citations to each of these scales during each year over the past two decades. We examined representative samples of 20 publications from 1974 to 1979 and 20 more from 1989 to 1994. Neck and neck over the years, the number of citations for each scale steadily increased. However, there was little growth in the concurrent use of both scales, and of these publications less than a quarter provided analyses of their differential outcomes. Reading the source publications showed little if any discussion of the rationale underlying the selection of these scales and little if any qualification of the overall results of the investigation as being-specific. Thompson's (1989) content analysis of depression scales by domains shows major differences between the Beck and the Hamilton. These differences are obscured in the articles reviewed. Each paper regards the scale they used as the gold standard without justifying its selection. We recommend that the particular symptoms that brought the patient to therapy should be considered in assessing clinical change. Clinimetric advances (Feinstein, 1987) have recognized that the measurement of change must be based on the clinical phenomena presented by the individual patient, rather than utter dependence on alterations in scores from psychometrically derived constructs.

T26. Attentional Biases In Peripersonal And Extrapersonal Space: An Analysis Of Line Bisection And Visual Orientation Performances In A Group Of Young Adults

Bruno J. Losier

In recent decades, attentional biases along three-dimensional spatial coordinates have been a topic of increased interest. However, physical boundaries, as well as methodological restraints have precluded a thorough and systematic investigations of human performances along these planes. The current study proposed to address some of these limitations by investigating line bisection and visual orientation performances along horizontal and vertical planes in peripersonal (within hand's reach) and extrapersonal (outside of hand's reach) space. Subjects bisected a total of 72 computer-generated horizontal and vertical lines projected in each spatial location. Furthermore, on any given trial, horizontal lines appeared at the top, centre or bottom of the screen, while vertical lines appeared at the left, centre or right of the screen. As well, a bisecting cursor appeared at either end of each line. In the visual orientation task, subjects responded to 229 validly or invalidly cued targets in each spatial location. On any given trial, targets appeared at the top left, top right, bottom left or bottom right of the screen. Overall, horizontal and vertical lines were bisected to the left and above the midpoint, respectively. Moreover, analyses of variance revealed that bisecting biases varied significantly in peripersonal and extrapersonal space contingent upon bisector and line position on the screen. Overall, subjects responded more quickly to validly than invalidly cued targets. Analyses of variance revealed that response time also varied significantly and differently along horizontal and vertical planes in peripersonal and extrapersonal space. These findings are discussed in light of current models of visual attention.

T27. O.C.E.A.N. - ology

T22 Michael Wm. MacGregor and Karina Davidson

The five-factor model is a taxonomy of personality traits in terms of five broad dimensions: Openness to Experience, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. Although evidence suggests that these five factors are useful in studying and modelling normal personality, there is less consensus regarding their ability to model abnormal personality, particularly personality disorders. This study investigated the relationship between the five factor model of personality and personality disorders.

Two hundred and fifty undergraduate students completed the Personality Diagnostic Questionnaire-Revised (PDQ-R) as a measure of personality disorders and the Big Five Inventory (BFI-54) as a measure of the five factors of personality. Results are discussed with respect to the theoretical relationship between personality disorders and the five factor model.

T28. Channel Surfing Without A Remote Control

Tracy L. Taylor(1), Dennis P. Phillips(1,2), & Susan E. Hall(1). Departments of Psychology(1) and Otolaryngology(2)

In an auditory gap detection task, three highly practiced listeners were presented with pairs of noise stimuli. Each stimulus consisted of leading and trailing noise bursts which bordered a temporal gap interval: In the test stimulus, this gap interval was varied systematically; in the reference stimulus this gap interval was maintained at a duration of 1 msec and was inaudible. The subject made a forced choice manual discrimination to indicate which of the two randomly presented stimuli contained the test gap. The dependent variable of interest was the shortest temporal gap that could be detected reliably. Subjects performed this task under conditions in which the perceptual channel used to define the leading and trailing edges of the gap was manipulated with respect to central frequency and laterality. In other words, perceptual channels were defined according to frequency and ear, such that the leading and trailing noise bursts shared no or complete spectral overlap and were presented to the same or to different ears. Results demonstrated poorer gap detection for across-channel (>20 msec) than for within-channel (< 5 msec) conditions, indicating that having to extract temporal information across frequency or ear limits the ability to resolve a temporal gap. It is noted that the thresholds observed in the across-channels conditions could be responsible for the +20 to +40 msec voice onset time (VOT) that typically marks the categorical boundary between voiced and voiceless stop consonants.

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no
end session 5

T29. Sex Differences: It's Not the Size, It's the Shape

Sick!

Margo Watt & Jeannette McGlone

As long ago as 1880, English surgeon James Crichton-Brown reported slight differences in the brain anatomy of men and women - a few more neurons *here* in one sex, a few more *there* in the other. But by far one of the most frequent findings through the years has been that the corpus callosum - the main fiber tract connecting the two cerebral hemispheres - tends to be larger in women than in men. It has also been found to vary in shape between the sexes. Researchers, such as deLacoste-Utamsing and Holloway (1982), have speculated that if a larger corpus callosum implies a larger number of fibers interconnecting cortical areas and the number of interhemispheric fibers correlates inversely with lateralization of function, then this finding would be consistent with neuropsychological theories that men's brains are more lateralized for function than women's. We review the last ten years of anatomical and radiological studies of sex differences in inter-hemispheric fibers pointing out inconsistencies in the methodologies employed. We conclude that certain segments of the corpus callosum linked to posterior association areas are greater in women. Whether these relate to functional asymmetries remains to be seen. Stay tuned!

T30. The Key Is In The Pee

Heather Schellinck and Richard Brown

Individual animals can be identified by their urine odours. To determine the influence of genetic, microbiological and dietary factors in producing urinary odours of individuality in rodents, a series of experiments were undertaken using two different paradigms. Results from a habituation-dishabituation task and a go/no go operant task revealed that rats and mice with genetic differences at the major histocompatibility complex have urine odours which can be discriminated by conspecifics. Bacteria are necessary for the production of these urine odours and the removal of different types of bacteria by drug treatments changes the nature of the urine odour; however, the specific nature of the odour-producing bacteria could not be determined. Mice and rats on different diets produce urine odours which are easier to discriminate than urine odours from mice of different MHC genotypes. These results indicate that odours of individuality are multidimensional and that when dietary cues are available, they may be more influential than genetic factors in providing discriminable cues.

T31. Imagine That

Anne Israeli and Gail Eskes

Previous research suggests that semantic processing using imagery reduces priming on implicit retrieval tasks of memory while increasing performance on explicit tasks. Eighty undergraduates were randomly assigned to study conditions involving the reading of words, imagery of the words, imagery of different words, and reading of words followed by reading different words. The implicit and explicit retrieval tests were word stem completion and cued recall, respectively. Preliminary results will be discussed.

end session 1p

P1. Infant "Surprise" Expressions as Coordinative Structures

Linda Lambrecht

Although infant "surprise" facial expressions involve opening the mouth and raising the eye-brows, these movements may represent movement synergies. Thirteen infants (5-7 months-old) were videotaped during toy presentations. Facial movements were coded for each infant. For target segments, the facial expressions were identified and coded using Ekman and Friesen's Facial Action Coding System. Two 4-sec target and two 4-sec control segments were coded while the infant's mouth was open (AU 26 or AU 27). All segments were coded for brow, eyelid, and nose movements. All infants produced no brow movements during control episodes. Mouth opening does recruit brow raises in nonemotional situations.

P2. Studies On The Expression Of Immediate-Early Genes (IEGs) In The Mammalian Circadian System

Mario E. Guido, Benjamin Rusak and Harold Robertson, Departments of Psychology and Pharmacology.

The mammalian suprachiasmatic nucleus (SCN) of the anterior hypothalamus is the site of an endogenous pacemaker that is responsible for the timing and overt expression of daily rhythms. Light-dark cycles synchronize circadian rhythms. The SCN receives 2 projections with visual information, one originating in the retina and the another one originating in a retinorecipient area of the lateral geniculate nucleus. A molecular correlate of photic entrainment is the rapid and transient induction of various IEGs in SCN cells. IEGs encode for a family of transcription factors such as Fos and Jun proteins; they may act as intracellular third messengers in the transduction of external signals to DNA. IEG proteins form complexes that bind to specific DNA sequences, promoting transcriptional activation of target genes. The pattern of IEG proteins induced by light is similar to the pattern of retinohypothalamic fiber terminals in the SCN. Fos and Jun proteins could be regulating transcriptionally the expression of genes responsible for the timing of intracellular circadian pacemaker.

In order to obtain more information about the role that IEGs could be playing in the SCN, we have tried to characterize the photic and circadian expression of 4 different IEGs: c-fos, fosB, c-jun and junB by in situ hybridization. At first, we studied the time course of expression of these genes in response to a light pulse, during the dark phase, when light has phase shifting effects. Second, we assessed the circadian expression of the mentioned genes at different circadian times, during the subjective day and night. JunB appears to play a special role at the beginning of subjective day.

P3. Orientational Singletons Evoke Facilitation As Measured Using Illusory Line Motion

Shelley McColl and Billy Schmidt

A growing body of research suggests that a highly salient featural singleton will only capture attention if the singleton is task-relevant (Yantis et al., ARVO '94). Such research, which uses visual search methodologies, assumes that the attentional capture is the result of a deliberate shift in attention towards the informative singleton. If the singletons are not informative, no facilitative advantage is said to be observed to such items. The present experiments probed for the existence of spatial facilitation around task-irrelevant orientational singletons using illusory line motion (Hikosaka et al., ARVO '91). Under task-irrelevant conditions, orientational singletons were found to produce facilitation as detected by illusory line motion. The observed difference in results between visual search and illusory line motion may result from the measurement of different types of facilitation by these two tasks.

P4. The Scent Of A Pregnant Woman: Rescent And Scentsitive Procedures For Making Scentse Out Of Changes In The Smell And Taste Scentses

Farhad Dastur¹, Richard Brown¹, Patrick McGrath¹, Donald Brown², Kathy Harrigan³ & Heather Wiles³ ¹Dept. of Psychology, ²Dept. of Family Medicine, ³Dietary Clinic, Grace Maternity Hospital

Women frequently report changes in odor and taste perception during pregnancy. Despite the potential such changes have in modifying food intake--thereby affecting nutritional health--there are a scarcity of comprehensive, well-designed studies examining the specific nature and direction of smell and taste alterations associated with pregnancy. In this prospective, controlled, longitudinal study, 64 women (32 pregnant subjects, 32 control subjects) will be assessed four times over a nine month period: (1) 6-10 weeks postconception (middle of first trimester); (2) 14-18 weeks postconception (middle of second trimester); (3) 28-32 weeks postconception (early to middle of third trimester); and (4) 4-8 weeks postpartum. Non-pregnant women will be tested at approximately equivalent intervals from the time of their first assessment and at consistent times during the early part of their menstrual cycle. Alterations in smell and taste will be assessed using a battery of objective psychophysical tests, questionnaires, rating scales, and diet records designed to measure the following: (a) odor and taste thresholds; (b) odor and taste identification; (c) qualitative ratings of odor and taste preference, familiarity, intensity, arousal, and palatability; (d) odor-evoked memories; (e) the development of food cravings and aversions; and (f) the preferences of the smells and tastes of non-food items. For the pregnant women, test battery results will be correlated with their serum hormone levels of estradiol, progesterone, human chorionic gonadotropin, oxytocin, and prolactin in an effort to reveal hormonal influences on the smell and taste senses. Additionally, the degree of pregnancy sickness during the first trimester will be assessed and will be correlated with performance on the perceptual tests. We hypothesize that changes in smell and taste perception during pregnancy may be the sensory mechanism underlying food cravings and aversions. We also hypothesize that during early pregnancy superior smell and taste performance should be positively correlated with the degree of severity of pregnancy sickness. These hypotheses are based on the theory that pregnancy sickness is a maternal evolutionary adaptation designed to avoid the ingestion of naturally occurring toxins, thus protecting the embryo from exposure to potentially harmful substances. A potential application of this research will be to help provide a rational basis for updating the nutritional guidelines available to pregnant women.

P5. Voluntary Finger Movements Can Be Controlled By The Supplementary Motor Area Without Participation Of The Primary Motor Cortex

C.D. MacKinnon and W. G. Tatton, Departments of Physiology and Physical Therapy, University of Toronto, and the Department of Psychology, Dalhousie University, Halifax, Nova Scotia

Prominent direct projections from the supplementary motor area (SMA) to the spinal cord in monkeys have been hypothesized to drive movements independently of the primary motor cortex (M1) (Dum and Strick, 1991). We tested this hypothesis by determining the location and time course of intracerebral activity time-locked to the generation of voluntary finger movements by collecting EEG for dipole source analysis (DSA) and positron emission tomography (PET) simultaneously. Experiments were conducted to directly compare unilateral serial, repetitive finger opposition movements (SMs) performed at 2 Hz to the same movements performed intermittently (IMs) at 0.2 to 0.08 Hz. EEG epochs were obtained during the IMs by back-averaging 2.5 sec prior to the onset of EMG activity and 0.5 after EMG onset. $H_2^{15}O$ PET scans were performed during the collection of the EEG epochs and foci co-registered on MRI and the DSA solutions. PET data demonstrated that the SMs involved significant activation of the contralateral M1 together with other structures in the cerebello-thalamocortical pathways and minimal activation of the SMA with minimal activation of M1. The SMA dipole orientation suggested that the IMs were generated by neuronal populations in the dorsal tier of the cingulate sulcus. We hypothesize that IMs are generated via basal ganglia-thalamocortical pathways which do not involve M1 and are separate from the pathways generating SMs. Both the SMs and IMs are presently being investigated in patients with Parkinson's disease and neuroleptic induced Parkinsonism. (C.D.M. supported by Ontario Ministry of Health Fellowship #04515).

P6. Object Recognition And Orientation: Basically Independent While Specifically Dependent

J.P. Hamm and Patricia McMullen

The time to name line drawings of objects rotated in the picture-plane show effects of orientation which support normalization of the image through the shortest angular distance from the upright (Jolicoeur, 1990) prior to identification. However, this notion is logically unsound (Corballis, 1988) in that the determination of the shortest direction requires information pertaining to the top of the object, which relies on knowing the identity of the object (Rock, 1973). How then is an image rotated for identification if the rotation process requires knowing the identity? The current research suggests a resolution to this conundrum lies in the level to which the object image must be identified. Using a word-picture verification task, Rosch, Mervis, Gray, Johnson, and Boyes-Braem, 1976 and Jolicoeur, Gluck, and Kosslyn (1984) have shown that objects are initially identified at a basic level (i.e., "dog") and that subordinate identification (i.e., "collie") requires further visual processing of the image. We report a word-picture verification task which extends these findings by showing that access to a basic level representation is orientation invariant, while access to the subordinate representation is sensitive to image orientation. These findings place the locus of the orientation effects in the visual processing required for subordinate access (Jolicoeur, et al, 1984). We speculate that these additional visual processes may include the processing the spatial relations of object parts which is known to be sensitive to stimulus orientation (McMullen and Jolicoeur, 1997; Kori et al and Norman, 1989).

P7. Effects of Dopaminergic D1, D2 and D3 Receptor Agonists and Antagonists on Body Temperature, Activity Level and the Emission of Ultrasonic Vocalizations by Rat Pups

Farhad Dastur¹, Richard E. Brown¹, Denise Wiles¹ and Iain McGregor², Departments of Psychology, ¹Dalhousie University ²University of Sydney, Sydney, Australia.

Infant rodents emit ultrasonic vocalizations (USVs) when they are cold, isolated from their littermates or roughly handled. As dopamine D1 receptor agonists induce hyperthermia, while D2 and D3 agonists induce hypothermia, we predicted that the D1 agonists would reduce USVs while D2 and D3 agonists would increase USVs in 11-13 day old rat pups. Pups were injected with drugs or saline and maintained at 33-36° C for 30 min before a 5 min USV test was conducted in a test chamber at 22-24° C. Body temperature (auxillary) was measured with a surface probe before and after USV recording. USVs were monitored by a QMC bat detector coupled to a computer and activity was recorded by an infrared detector. During the 5 min test, the body temperature of the pups given the high dose (2.5 mg/kg) of the D1 agonist SKF 81297 was unexpectedly lower than the controls and the pups emitted fewer USVs and had decreased activity levels. As SKF increases body temperature in adults, the reduced body temperature in pups was unexpected. The low dose of SKF 81297 (0.25 mg/kg) had no significant effect on body temperature, USVs or activity. SCH 23390 (0.3 mg/kg) but not Raclopride (0.3 mg/kg) reversed the effect on temperature. The D2 agonist Quinpirole caused a significant decrease in body temperature at both low (0.3 mg/kg) and high (3.0 mg/kg) doses and both doses increased activity levels, but only the high dose reduced USVs. These effects were reversed by Raclopride, but not SCH 23390. The D3 agonist 7-OH-DPAT reduced temperature at the high dose (1.0 mg/kg) but not at the low dose (0.1 mg/kg). Both doses increased activity levels and this effect was reversed by both antagonists. Only the high dose of 7-OH-DPAT reduced USVs and this effect was reversed by Raclopride but not by SCH 23390. Neither the D1 antagonist SCH 23390 (0.3 mg/kg) nor the D2 antagonist Raclopride (0.3 mg/kg) given alone altered body temperature or USVs, but SCH 23390 reduced activity levels. The D1, D2 and D3 agonists all reduced USVs at high doses and, across all groups, as body temperature dropped, there were fewer USVs, but higher activity levels. These results indicate that the drop in body temperature during the 5 min test was not the most important stimulus for emission of USVs in 11-13 day old rats. The overall reduction in USVs by dopamine agonists suggests that the antianxiety or antidepressant actions of these drugs, their reinforcing properties, or their effects on respiration alter USV emission in infant rats, rather than their thermoregulatory actions.

P8. Bill Chaplin

An Alternative Approach to Statistical Inference: Rejecting a Null Hypothesis is as Boring as this ~~Talk~~ Title

Some thoughts on an alternative approach to statistical inference will be presented.

P9. Initial Steps Toward A Model Of Numerosity Discrimination In The Pigeon

W. R. Matheson, Simon Gadbois, and W. K. Honig

A series of studies has shown that pigeons can estimate the relative and absolute numerosities of arrays of small elements, and can discriminate related properties such as uniformity. These abilities have been demonstrated with both successive and simultaneous discrimination procedures, and with elements which differed on dimensions including color, shape, and size. We have begun to develop an information processing model to provide a framework to explain these results, and, more specifically, to analyze relative numerosity discriminations as secondary to concurrent processes of absolute discrimination. Our model is based on the work of J. Gibbon, R. Church, and C. R. Gallistel, particularly on Gibbon's scalar timing theory. We are also using Monte Carlo simulations to derive predicted relative numerosity gradients from data obtained during absolute numerosity tasks. In future research, the collection of trial by trial data will assist in narrowing the range of possible models.

Symposium

Where Are We Going? Psychology and Neuroscience in the 21st Century

300 Chair: Richard Brown

315 **Whatever's To Become of Us?**
Jim Clark

Well...now see...ummm...but...uh...so...gee...hmmm...then there's...duh...er...eh?

330 **Can Behavioural Neuroscience Continue Without Incorporating Cellular and Molecular Neuroscience?**
Bill Tatton

345 **Towards an Integrative Neuroscience**
John Fentress

400 **Prospects for Cognitive Neuroscience in the 21st Century**
Ray Klein

475 **Building Persons: A Goal for the 21st Century?**
John Barresi

495 **Shoot From the Hip Cowboys, True Believers, or Boring Scientists: The Future of Clinical Psychology**
Patrick McGrath

Clinical psychology is at a crossroads or maybe it is a cloverleaf that will determine if we survive as a legitimate endeavor that improves the health of Canadians or if we become a marginalized group of academics and practitioners. There are three archetypes of clinical psychology that are coming to the forefront: Shoot from the Hip Cowboys (S.H.C.'ers); True Believers (T.B.'ers) and Boring Scientists (B.S.'ers). I will outline the strengths and weaknesses of each archetype and make foolhardy predictions about their future.

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what are the top 20 / top 100 terms that we think psychology students should learn?

1. Defn of Psychology
2. Top 100 terms